



Fashion Retail Sales Analysis

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Executive summary

This report presents a detailed exploratory data analysis of a fashion retail company's sales transactions. This analysis uses Python's data science stack to uncover critical insights into the brand's sales performance, their customers' behaviors, product popularity and purchasing trends.

Key findings from this study indicate strong revenue generation with identifiable seasonal trends, a healthy base of concurrent customers, and important chances for strategic growth in specific product categories and genres, and of course, certain customer audiences.

The insights resulting from this analysis are intended to inform data-driven decision-making for mainly marketing, product creation and customer-brand relationship strategies.

1. Introduction

1.1 Objectives

- Transform raw data into actionable results by identifying patterns and key performance indicators.
- Generate valuable visual information from the main dataset, portraying easily detectable trends from charts, histograms, etc.
- Gain new information from the insights that may be helpful to the company's strategic development.

1.2 Dataset & Tools

This analysis was conducted on a cleaned dataset containing over 2000 transactions. The dataset contains the following fields:

- customer_id
- item_purchased
- purchase_amount_usd
- date_purchase
- review_rating
- payment_method
- purchase_year
- purchase_month
- purchase_quarter

1.3 Dataset overview

- Total transactions: 2750
- Period of data: 2022-10-02 (00:00:00) to 2023-10-01 00:00:00
- Total of unique clients: 166
- Total of unique products: 50
- Price range: \$10.00 - \$4932.00

2. Results

2.1 Descriptive Analysis

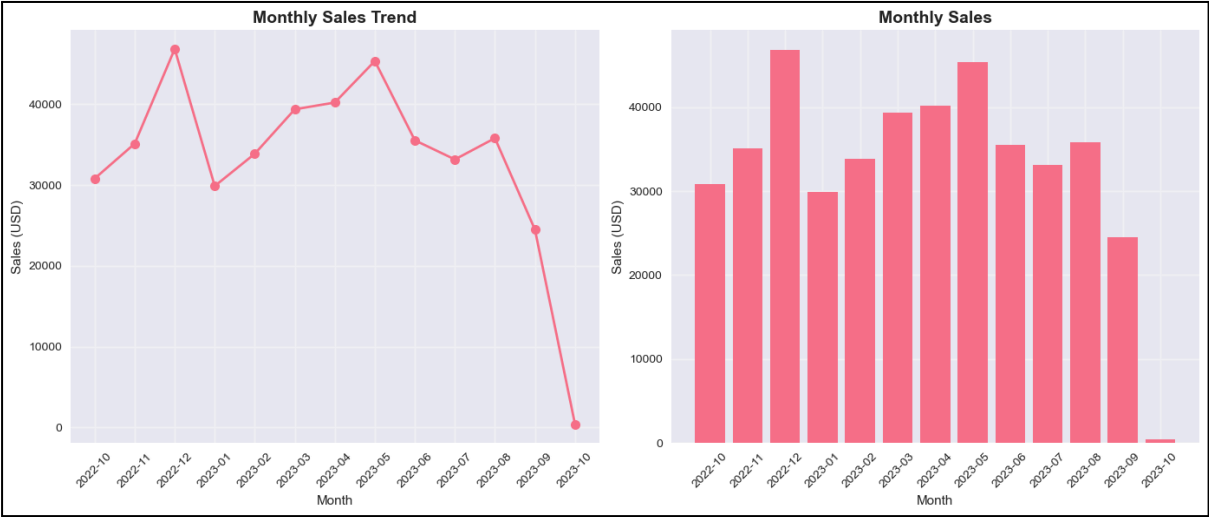
Out of all the possible variables we can study in a descriptive way, there are only two which are significant to the final conclusions: **purchase_amount_usd** and **review_rating**. To have a general knowledge about their behavior on all the customers, we have to know their *count*, *mean*, *std*, *min*, *quarters* and *maximum* values.

	purchase_amount_usd	review_rating
count	2750.000000	2487.000000
mean	156.709818	2.986248
std	419.536669	1.150691
min	10.000000	1.000000
25%	57.000000	2.000000
50%	110.000000	3.000000
75%	155.750000	4.000000
max	4932.000000	5.000000

2. 2 Sales Analysis

We are having a look at the monthly sales that the brand has had during the studied period of data (2022-10-02 (00:00:00) to 2023-10-01 00:00:00).

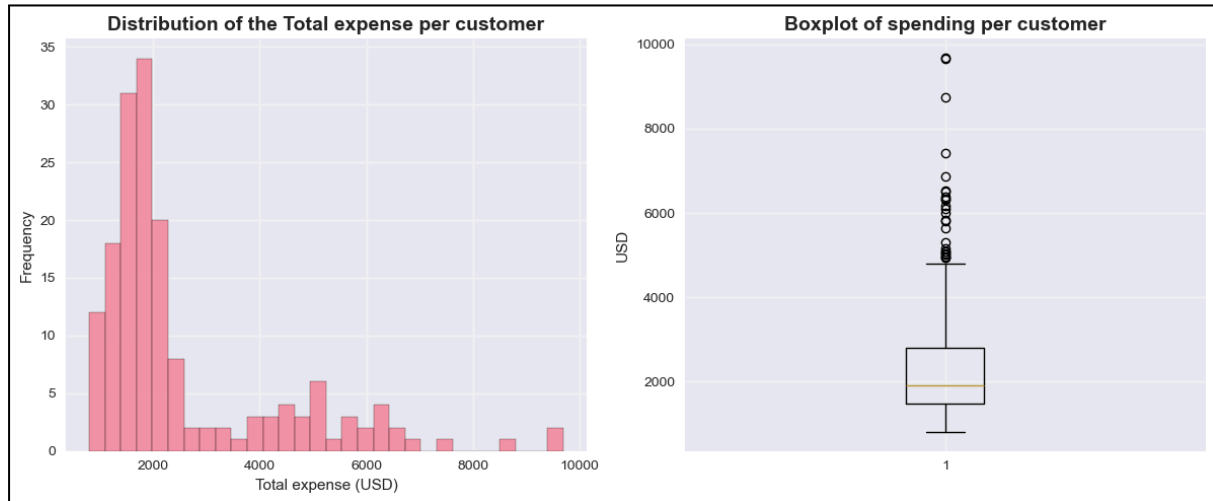
purchase_year	purchase_month	total_sales	transaction_count	avg_transaction
2022	10	30812.0	246	125.25
	11	35129.0	198	177.42
	12	46851.0	248	188.92
2023	1	29887.0	237	126.11
	2	33879.0	198	171.11
	3	39380.0	232	169.74
	4	40220.0	235	171.15
	5	45352.0	250	181.41
	6	35542.0	209	170.06
	7	33176.0	246	134.86
	8	35809.0	248	144.39
	9	24492.0	198	123.70
	10	423.0	5	84.60



2.3 Client Analysis

Client statistics:

- Customers with one purchase only: 0
- Recurrent customers: 166
- Customer that has spent the most: \$9685.00
- Average expense of customers: \$2596.10



2.4 Product Analysis

Top 10 of most sold products:

item_purchased	units
Shorts	74
Pants	71
Skirt	70
Belt	69
Tank top	66
T-shirt	63
Camisole	62
Hoodie	62
Cardigan	61
Pajamas	61

Top 10 products per revenue

item_purchased	revenue	units_sold
Tunic	17275.0	54
Jeans	13068.9	48
Pajamas	12798.0	61
Shorts	12702.0	74
Hangbag	12668.0	59
Gloves	12330.0	52
Boots	11891.0	52
Poncho	11422.0	60
Flip-flops	11309.0	41
Slippers	11284.0	46

2.5 Payment methods analysis

Statistics by payment method:

payment_method	transaction_count	total_amount	avg_amount	avg_rating
cash	1312	200338.0	152.70	2.98
credit card	1438	230614.0	160.37	2.99

2.5 Seasons analysis

Sales by day of the week:

day_of_week	sum	count
Friday	70759.0	392
Monday	64662.0	412
Saturday	52175.0	392
Sunday	71497.0	378
Thursday	57341.0	429

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Tuesday	61722.0	372
Wednesday	52796.0	375

- By looking at this table, we observe that Thursday has sold the most units out of the week's days so far, but still isn't the day that most revenue has generated. That can be interpreted as a fact that on Thursday, customers may tend to buy basic clothing rather than expensive choices.

On the other hand, Sunday generates higher revenue with fewer units sold compared to Thursday, suggesting that customers purchase more expensive items on that day.

Friday also shows strong sales both in revenue and units, which may be linked to customers buying slightly more expensive items in preparation for parties, events, or weekend activities.

Sales by month:

month_name	sum	count
April	40220.0	235
August	35809.0	248
December	46851.0	248
February	33879.0	198
January	29887.0	237
July	33176.0	246
June	35542.0	209
March	39380.0	232
May	45352.0	250
November	35129.0	198
October	31235.0	251
September	24492.0	198

- Highest revenue (sum): December (46851 units sold), followed closely by May (45352)
- Lowest revenue → September (24492)
- Most units sold (count) → October (251), but not with the highest revenue
- Least units sold: February, November and September (all 198)

Interpretation:

- October** has the highest number of sales but not high revenue, which indicates that customers have bought cheaper items on the majority.
- December** leads in revenue, likely due to holiday/Christmas shopping, despite not having the highest unit count, which indicates that customers have bought more expensive items.

- **May** is also a strong revenue month, close to December, and with a high unit count too, which indicates balanced strength (units + value).
- **September** has the lowest revenue despite not being the absolute lowest in unit count. It could be a low demand month with less shopping activity.

3. Conclusions

From the data and analysis done, there are 3 key conclusions we can mark:

1. **Strong Seasonal Performance with Clear Peaks**
 - **December** emerges as the highest revenue month with the amount of \$46,851, influenced by holiday shopping and New Year's Eve events.
 - **May** shows a unique performance with both high revenue: An amount of \$45,352 sold and second month with the most units sold (250 units)
 - **September** is the lowest performing month, indicating a need for targeted promotions during this period
2. **Customer Behavior Patterns**
 - **Solid average customer** value of \$2596.10, indicating strong brand engagement and a loyal brand-customer relationship.
3. **Product Portfolio Insights**
 - **High-volume** basics dominate in unit sales.
 - **Premium categories drive revenue** despite selling moderate quantities of units.
 - **Opportunity for margin optimization**